# SANTOSH KUMAR

Systems & Search Engine Architect · Full-Stack & Native Mobile Engineer

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## EXECUTIVE PROFILE

Senior Systems & Mobile Software Engineer specializing in low-latency in-memory search systems, cryptographic authentication engines, and high-framerate mobile platforms. Creator of SearchHub and DocPulse (14KB zero-dependency in-browser lexical search cores with Damerau-Levenshtein typo correction) and Dev Showcase (120fps native Flutter platform following Clean Architecture and Riverpod state management). Proven track record in deterministic state machines, automated CI/CD pipelines, and robust production hardening.

## TECHNICAL COMPETENCIES & ARCHITECTURAL SKILLS

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| **Languages & Runtimes:** | TypeScript, JavaScript (ESNext), Dart, Node.js, SQL, HTML5/CSS3 |
| **Frameworks & Libraries:** | React 19, Next.js (App Router), Flutter, Tailwind CSS, Vite, Riverpod |
| **Search & Algorithms:** | BM25 Lexical Ranking, TF Saturation, Inverted Indexes, Damerau-Levenshtein Distance |
| **Security & Cryptography:** | Scrypt KDF, PBKDF2, Web Crypto API, XSS Protection (DOM TextContent), CSP |
| **Architecture & DevOps:** | Domain-Driven Design (DDD), Clean Architecture, GitHub Actions CI/CD, Fastly CDN Edge |

## FLAGSHIP PRODUCTION REPOSITORIES & OPEN-SOURCE WORK

### SearchHub & SearchEmbed (DocPulse v1.0.0)

github.com/santoshkumarvvv/search-hub- | search-hub-pearl.vercel.app

*Lead Architect & Engineer | TypeScript, BM25 Lexical Search Core, Vite, Next.js*

- Engineered a 14.8KB zero-dependency in-browser lexical search widget replacing heavy Algolia and Elasticsearch setups.
- Implemented client-side inverted indexing with Term-Frequency (TF) saturation and bounded Damerau-Levenshtein typo correction (tokens >= 4 chars).
- Designed an accessible ARIA modal dialog with complete focus-trapping (Tab/Shift+Tab), Esc dismissal, trigger focus restoration, and pure DOM textContent rendering to eliminate XSS risks completely.

### Dev Showcase Mobile Platform

github.com/santoshkumarvvv/dev-showcase

*Creator & Mobile Architect | Flutter, Dart, Riverpod, Clean Architecture*

- Developed a 120fps smooth native Android/iOS showcase application implementing strict Clean Architecture with Data, Domain, and Presentation separation.
- Utilized Riverpod for deterministic reactive state management and immutable view-model transformations.
- Integrated automated GitHub Actions CI/CD workflows for linting, test suites, and multi-architecture release artifact compilation.

## ADDITIONAL SYSTEMS & MOBILE PROJECTS

### SearchHub Mobile Companion App

github.com/santoshkumarvvv/search-hub-app

*Full-Stack Mobile Engineer | Flutter, RESTful Edge APIs, Offline Cache*

- Built native mobile extension enabling real-time documentation indexing and cross-platform search federation.
- Engineered offline cache layers with SQLite and in-memory LRU eviction policies ensuring sub-second response times in zero-connectivity environments.

### Scrypt KDF Cryptographic Engine & Security Architecture

santoshkumarvvv.github.io/#works

*Security & Systems Engineer | Cryptography, Web Crypto API, Inverted Index*

- Engineered sequential memory-hard key derivation functions (N=16384, r=8, p=1) defending against specialized ASIC and GPU brute-force attacks.
- Enforced rigorous Content Security Policies (CSP) and zero-eval JavaScript execution standards to block cross-site scripting.

## ENGINEERING METHODOLOGY & PRODUCTION STANDARDS

**Zero Marketing Fiction:**
Every benchmark, algorithm claim, and deployment status is grounded in reproducible metrics. Zero mock APIs, zero hardcoded shells.

**Zero-Dependency Micro-Cores:**
Designing client embed widgets under 15KB gzipped to minimize bundle overhead, cold-start latency, and external supply-chain vulnerabilities.

**Universal Accessibility & SEO:**
Complete WCAG AAA contrast, semantic HTML prerendering, Schema.org JSON-LD structured data, and keyboard-first navigation patterns.

## EDUCATION & VERIFIED CREDENTIALS

### Bachelor of Technology / Computer Science & Systems Engineering

India

Focus: Distributed Systems, Cryptography, Algorithm Analysis & Mobile Architecture

## DIRECT CONTACT & COLLABORATION INQUIRIES

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